

Lines, Triangles & Polygons

ANSWER KEY



Parallel lines and transversals

- 1 Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.

Corresponding angle = 105° (equal). Alternate interior angle = 105° (equal). Co-interior angle = $180^\circ - 105^\circ = 75^\circ$ (supplementary).

- 2 Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.

$3x + 10 = 5x - 30 \Rightarrow 40 = 2x \Rightarrow x = 20$. The angle measures $3(20) + 10 = 70^\circ$.

- 3 Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.

Corresponding angles are equal: $4x + 15 = 6x - 25 \Rightarrow 2x = 40 \Rightarrow x = 20$. The angle measures $4(20) + 15 = 95^\circ$.

Triangle angle sum and the exterior angle theorem

- 4 The angles of a triangle are in the ratio $3 : 5 : 7$. Find each angle.

The parts sum to $3 + 5 + 7 = 15$, and the total is 180° , so each part is 12° . The angles are 36° , 60° , and 84° .

- 5 In triangle ABC , $\angle A = 52^\circ$ and the exterior angle at C is 130° . Find $\angle B$.

The exterior angle equals the sum of the remote interior angles: $130^\circ = 52^\circ + \angle B \Rightarrow \angle B = 78^\circ$.

- 6 An exterior angle of a triangle measures 115° , and one remote interior angle is 50° . Find the other remote interior angle and the third (adjacent) interior angle.

The other remote interior angle is $115^\circ - 50^\circ = 65^\circ$. The adjacent interior angle is supplementary to the exterior angle: $180^\circ - 115^\circ = 65^\circ$. Check: $50^\circ + 65^\circ + 65^\circ = 180^\circ$.

Triangle congruence criteria

- 7 State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.

SAS — two sides and the included angle.

- 8 Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?

Yes — congruent by SAS, since two sides and the included right angle are equal in both triangles.

- 9 Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?

SSS applies, and yes, the triangles are necessarily congruent.

Similar triangles and proportional reasoning

- 10 Triangle $ABC \sim$ triangle DEF , with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF .

$$\text{Scale factor} = \frac{DE}{AB} = \frac{9}{6} = 1.5. \quad EF = 8 \times 1.5 = 12.$$

- 11 Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF , where $DE = 15$ corresponds to AB . Find EF and FD .

$$\text{Scale factor} = \frac{DE}{AB} = \frac{15}{12} = 1.25. \quad EF = BC \times 1.25 = 16 \times 1.25 = 20. \\ FD = CA \times 1.25 = 20 \times 1.25 = 25.$$

- 12 Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK . Given $MN = 10$, verify the scale factor is consistent, then find NO .

$$\text{Check: } \frac{MN}{JK} = \frac{10}{15} = \frac{2}{3}, \text{ consistent. } NO = KL \times \frac{2}{3} = 21 \times \frac{2}{3} = 14.$$

The Law of Cosines

- 13 In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.

$$x^2 = 9^2 + 13^2 - 2(9)(13)\cos 50^\circ \approx 250 - 150.41 = 99.59, \text{ so } x \approx \sqrt{99.59} \approx 9.98.$$

- 14 A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side C), to the nearest tenth of a degree.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 64 - 100}{112} = \frac{13}{112} \approx 0.1161. \quad C \approx 83.3^\circ.$$

- 15 A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c , to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.

$$c^2 = 6^2 + 6^2 - 2(6)(6)\cos 80^\circ = 72 - 72(0.1736) \approx 59.50, \text{ so } c \approx 7.71. \text{ Since two sides are equal, the triangle is isosceles.}$$

Heron's formula

- 16** Find the area of a triangle with sides 13, 14, 15 using Heron's formula.

$$s = \frac{13 + 14 + 15}{2} = 21. \text{ Area} = \sqrt{21(21 - 13)(21 - 14)(21 - 15)} = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84.$$

- 17** A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

$$s = \frac{9 + 10 + 17}{2} = 18. \text{ Area} = \sqrt{18(18 - 9)(18 - 10)(18 - 17)} = \sqrt{18 \cdot 9 \cdot 8 \cdot 1} = \sqrt{1296} = 36.$$

- 18** A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).

$$s = \frac{5 + 12 + 13}{2} = 15. \text{ Area} = \sqrt{15 \cdot 10 \cdot 3 \cdot 2} = \sqrt{900} = 30 \text{ m}^2. \text{ Check: since it's a right triangle with legs 5 and 12, Area} = \frac{1}{2}(5)(12) = 30 \text{ m}^2. \text{ Matches.}$$

Interior and exterior angles of polygons

- 19** Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.

$$\text{Sum} = (10 - 2) \times 180^\circ = 1440^\circ. \text{ Each interior angle} = \frac{1440^\circ}{10} = 144^\circ.$$

- 20** Each exterior angle of a regular polygon measures 24° . How many sides does the polygon have?

$$n = \frac{360^\circ}{24^\circ} = 15 \text{ sides.}$$

- 21** A regular polygon has an interior angle of 150° . How many sides does it have?

$$\text{Exterior angle} = 180^\circ - 150^\circ = 30^\circ. n = \frac{360^\circ}{30^\circ} = 12 \text{ sides.}$$

Area of polygons

- 22** The trapezoid below has parallel sides of length 14 and 8, and height 6. Find its area.

$$\text{Area} = \frac{1}{2}(14 + 8)(6) = \frac{1}{2}(22)(6) = 66 \text{ square units.}$$

- 23** Find the area of a regular hexagon with side length 6.

$$\text{Area} = \frac{3\sqrt{3}}{2}s^2 = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3} \approx 93.5 \text{ square units.}$$

- 24** A composite figure is an 8-by-10 rectangle with a 3-by-4 rectangular notch removed from one corner. Find the remaining area.

$$\text{Rectangle area} = 8 \times 10 = 80. \text{ Notch area} = 3 \times 4 = 12. \text{ Remaining area} = 80 - 12 = 68 \text{ square units.}$$

A well-designed word problem: a surveyor's triangular lots

- 25** This question has three parts. A surveyor measures a triangular lot with sides 30 m, 40 m, and 50 m. (a) Verify using the Pythagorean theorem that the lot is a right triangle. (b) Find the area of the lot using the right-triangle area formula, then confirm your answer using Heron's formula. (c) A second, similar triangular lot has its shortest side measuring 45 m. Using the similarity ratio, find the area of the second lot.

(a) $30^2 + 40^2 = 900 + 1600 = 2500 = 50^2$, confirming a right angle between the 30 m and 40 m sides.
(b) Right-triangle area $= \frac{1}{2}(30)(40) = 600 \text{ m}^2$. Heron's: $s = 60$, Area $= \sqrt{60 \cdot 30 \cdot 20 \cdot 10} = \sqrt{360000} = 600 \text{ m}^2$. Matches. (c) Similarity ratio $k = \frac{45}{30} = 1.5$. Area scales as $k^2 = 2.25$, so the second lot's area $= 600 \times 2.25 = 1350 \text{ m}^2$.

Multiple choice — easy

- 26** The angles of a triangle are in the ratio 2 : 1 : 3. Find each angle.

- a. 30, 64, 86
- b. 30, 60, 90 ✓
- c. 32, 60, 88
- d. 90, 60, 30

The parts sum to $2 + 1 + 3 = 6$, and the total is 180° , so each part is 30° . The angles are 30° , 60° , and 90° .

- 27** The angles of a triangle are in the ratio 5 : 9 : 6. Find each angle.

- a. 45, 58, 77
- b. 81, 54, 45
- c. 45, 54, 81 ✓
- d. 47, 54, 79

The parts sum to $5 + 9 + 6 = 20$, and the total is 180° , so each part is 9° . The angles are 45° , 54° , and 81° .

28 The angles of a triangle are in the ratio 9 : 3 : 3. Find each angle.

- a. 38, 36, 106
- b. 36, 40, 104
- c. 36, 36, 108 ✓
- d. 108, 36, 36

The parts sum to $9 + 3 + 3 = 15$, and the total is 180° , so each part is 12° . The angles are 36° , 36° , and 108° .

29 The angles of a triangle are in the ratio 6 : 5 : 7. Find each angle.

- a. 50, 64, 66
- b. 52, 60, 68
- c. 50, 60, 70 ✓
- d. 70, 60, 50

The parts sum to $6 + 5 + 7 = 18$, and the total is 180° , so each part is 10° . The angles are 50° , 60° , and 70° .

30 The angles of a triangle are in the ratio 1 : 5 : 6. Find each angle.

- a. 90, 75, 15
- b. 15, 79, 86
- c. 15, 75, 90 ✓
- d. 17, 75, 88

The parts sum to $1 + 5 + 6 = 12$, and the total is 180° , so each part is 15° . The angles are 15° , 75° , and 90° .

31 The angles of a triangle are in the ratio 2 : 3 : 5. Find each angle.

- a. 38, 54, 88
- b. 36, 58, 86
- c. 36, 54, 90 ✓
- d. 90, 54, 36

The parts sum to $2 + 3 + 5 = 10$, and the total is 180° , so each part is 18° . The angles are 36° , 54° , and 90° .

32 The angles of a triangle are in the ratio 4 : 5 : 6. Find each angle.

- a. 50, 60, 70
- b. 48, 64, 68
- c. 48, 60, 72 ✓
- d. 72, 60, 48

The parts sum to $4 + 5 + 6 = 15$, and the total is 180° , so each part is 12° . The angles are 48° , 60° , and 72° .

33 The angles of a triangle are in the ratio 5 : 8 : 2. Find each angle.

- a. 24, 60, 96 ✓
- b. 24, 64, 92
- c. 26, 60, 94
- d. 96, 60, 24

The parts sum to $5 + 8 + 2 = 15$, and the total is 180° , so each part is 12° . The angles are 24° , 60° , and 96° .

34 The angles of a triangle are in the ratio 9 : 2 : 9. Find each angle.

- a. 81, 81, 18
- b. 18, 81, 81 ✓
- c. 20, 81, 79
- d. 18, 85, 77

The parts sum to $9 + 2 + 9 = 20$, and the total is 180° , so each part is 9° . The angles are 18° , 81° , and 81° .

35 The angles of a triangle are in the ratio 3 : 3 : 9. Find each angle.

- a. 108, 36, 36
- b. 38, 36, 106
- c. 36, 40, 104
- d. 36, 36, 108 ✓

The parts sum to $3 + 3 + 9 = 15$, and the total is 180° , so each part is 12° . The angles are 36° , 36° , and 108° .

36 The angles of a triangle are in the ratio 2 : 3 : 1. Find each angle.

- a. 30, 64, 86
- b. 90, 60, 30
- c. 32, 60, 88
- d. 30, 60, 90 ✓

The parts sum to $2 + 3 + 1 = 6$, and the total is 180° , so each part is 30° . The angles are 30° , 60° , and 90° .

37 The angles of a triangle are in the ratio 9 : 5 : 6. Find each angle.

a. 45, 54, 81 ✓

b. 81, 54, 45

c. 47, 54, 79

d. 45, 58, 77

The parts sum to $9 + 5 + 6 = 20$, and the total is 180° , so each part is 9° . The angles are 45° , 54° , and 81° .

38 The angles of a triangle are in the ratio 1 : 2 : 6. Find each angle.

a. 22, 40, 118

b. 20, 40, 120 ✓

c. 120, 40, 20

d. 20, 44, 116

The parts sum to $1 + 2 + 6 = 9$, and the total is 180° , so each part is 20° . The angles are 20° , 40° , and 120° .

39 The angles of a triangle are in the ratio 5 : 6 : 1. Find each angle.

a. 15, 75, 90 ✓

b. 17, 75, 88

c. 90, 75, 15

d. 15, 79, 86

The parts sum to $5 + 6 + 1 = 12$, and the total is 180° , so each part is 15° . The angles are 15° , 75° , and 90° .

40 The angles of a triangle are in the ratio 8 : 3 : 4. Find each angle.

- a. 96, 48, 36
- b. 38, 48, 94
- c. 36, 48, 96 ✓
- d. 36, 52, 92

The parts sum to $8 + 3 + 4 = 15$, and the total is 180° , so each part is 12° . The angles are 36° , 48° , and 96° .

Multiple choice — medium

41 An exterior angle of a triangle measures 82° , and one remote interior angle is 30° . Find the other remote interior angle.

- a. 52° ✓
- b. 57°
- c. 47°
- d. 30°

The exterior angle equals the sum of the remote interior angles: $82^\circ = 30^\circ + \angle B \Rightarrow \angle B = 52^\circ$.

42 An exterior angle of a triangle measures 90° , and one remote interior angle is 39° . Find the other remote interior angle.

- a. 56°
- b. 46°
- c. 51° ✓
- d. 39°

The exterior angle equals the sum of the remote interior angles: $90^\circ = 39^\circ + \angle B \Rightarrow \angle B = 51^\circ$.

- 43 An exterior angle of a triangle measures 67° , and one remote interior angle is 25° . Find the other remote interior angle.

a. 42° ✓

b. 25°

c. 37°

d. 47°

The exterior angle equals the sum of the remote interior angles: $67^\circ = 25^\circ + \angle B \Rightarrow \angle B = 42^\circ$.

- 44 An exterior angle of a triangle measures 76° , and one remote interior angle is 44° . Find the other remote interior angle.

a. 27°

b. 44°

c. 37°

d. 32° ✓

The exterior angle equals the sum of the remote interior angles: $76^\circ = 44^\circ + \angle B \Rightarrow \angle B = 32^\circ$.

- 45 An exterior angle of a triangle measures 116° , and one remote interior angle is 67° . Find the other remote interior angle.

a. 54°

b. 49° ✓

c. 44°

d. 67°

The exterior angle equals the sum of the remote interior angles: $116^\circ = 67^\circ + \angle B \Rightarrow \angle B = 49^\circ$.

- 46 An exterior angle of a triangle measures 56° , and one remote interior angle is 28° . Find the other remote interior angle.

a. 28° ✓

b. 31°

c. 23°

d. 33°

The exterior angle equals the sum of the remote interior angles: $56^\circ = 28^\circ + \angle B \Rightarrow \angle B = 28^\circ$.

- 47 An exterior angle of a triangle measures 92° , and one remote interior angle is 48° . Find the other remote interior angle.

a. 49°

b. 48°

c. 44° ✓

d. 39°

The exterior angle equals the sum of the remote interior angles: $92^\circ = 48^\circ + \angle B \Rightarrow \angle B = 44^\circ$.

- 48 An exterior angle of a triangle measures 54° , and one remote interior angle is 23° . Find the other remote interior angle.

a. 23°

b. 36°

c. 31° ✓

d. 26°

The exterior angle equals the sum of the remote interior angles: $54^\circ = 23^\circ + \angle B \Rightarrow \angle B = 31^\circ$.

- 49 An exterior angle of a triangle measures 94° , and one remote interior angle is 67° . Find the other remote interior angle.
- a. 67°
 - b. 27° ✓
 - c. 32°
 - d. 22°

The exterior angle equals the sum of the remote interior angles: $94^\circ = 67^\circ + \angle B \Rightarrow \angle B = 27^\circ$.

- 50 An exterior angle of a triangle measures 95° , and one remote interior angle is 26° . Find the other remote interior angle.
- a. 26°
 - b. 69° ✓
 - c. 64°
 - d. 74°

The exterior angle equals the sum of the remote interior angles: $95^\circ = 26^\circ + \angle B \Rightarrow \angle B = 69^\circ$.

- 51 An exterior angle of a triangle measures 88° , and one remote interior angle is 22° . Find the other remote interior angle.
- a. 22°
 - b. 71°
 - c. 61°
 - d. 66° ✓

The exterior angle equals the sum of the remote interior angles: $88^\circ = 22^\circ + \angle B \Rightarrow \angle B = 66^\circ$.

- 52 An exterior angle of a triangle measures 51° , and one remote interior angle is 31° . Find the other remote interior angle.
- a. 31°
 - b. 20° ✓
 - c. 25°
 - d. 15°

The exterior angle equals the sum of the remote interior angles: $51^\circ = 31^\circ + \angle B \Rightarrow \angle B = 20^\circ$.

- 53 An exterior angle of a triangle measures 86° , and one remote interior angle is 54° . Find the other remote interior angle.
- a. 32° ✓
 - b. 54°
 - c. 27°
 - d. 37°

The exterior angle equals the sum of the remote interior angles: $86^\circ = 54^\circ + \angle B \Rightarrow \angle B = 32^\circ$.

- 54 An exterior angle of a triangle measures 89° , and one remote interior angle is 21° . Find the other remote interior angle.
- a. 21°
 - b. 68° ✓
 - c. 73°
 - d. 63°

The exterior angle equals the sum of the remote interior angles: $89^\circ = 21^\circ + \angle B \Rightarrow \angle B = 68^\circ$.

- 55 An exterior angle of a triangle measures 47° , and one remote interior angle is 25° . Find the other remote interior angle.
- a. 25°
 - b. 22° ✓
 - c. 27°
 - d. 17°

The exterior angle equals the sum of the remote interior angles: $47^\circ = 25^\circ + \angle B \Rightarrow \angle B = 22^\circ$.

Multiple choice — hard

- 56 A triangle has sides $a = 6$, $b = 11$, and included angle $C = 110^\circ$. Find side c , to the nearest hundredth.
- a. 13.72
 - b. 14.72
 - c. 14.22 ✓
 - d. 15.22

$$c^2 = 6^2 + 11^2 - 2(6)(11)\cos 110^\circ \approx 157 + 45.15 \approx 202.15, \text{ so } c \approx 14.22.$$

- 57 A triangle has sides $a = 12$, $b = 10$, and included angle $C = 40^\circ$. Find side c , to the nearest hundredth.
- a. 8.26
 - b. 8.76
 - c. 7.26
 - d. 7.76 ✓

$$c^2 = 12^2 + 10^2 - 2(12)(10)\cos 40^\circ \approx 244 - 183.85 \approx 60.15, \text{ so } c \approx 7.76.$$

58 A triangle has sides $a = 7$, $b = 14$, and included angle $C = 110^\circ$. Find side c , to the nearest hundredth.

- a. 18.16
- b. 17.16
- c. 18.66
- d. 17.66 ✓

$$c^2 = 7^2 + 14^2 - 2(7)(14)\cos 110^\circ \approx 245 + 67.04 \approx 312.04, \text{ so } c \approx 17.66.$$

59 A triangle has sides $a = 7$, $b = 13$, and included angle $C = 80^\circ$. Find side c , to the nearest hundredth.

- a. 13.65 ✓
- b. 14.15
- c. 14.65
- d. 13.15

$$c^2 = 7^2 + 13^2 - 2(7)(13)\cos 80^\circ \approx 218 - 31.60 \approx 186.40, \text{ so } c \approx 13.65.$$

60 A triangle has sides $a = 10$, $b = 6$, and included angle $C = 70^\circ$. Find side c , to the nearest hundredth.

- a. 10.74
- b. 9.74 ✓
- c. 9.24
- d. 10.24

$$c^2 = 10^2 + 6^2 - 2(10)(6)\cos 70^\circ \approx 136 - 41.04 \approx 94.96, \text{ so } c \approx 9.74.$$

61 A triangular plot of land has sides 12, 35, 37. Find its area using Heron's formula.

- a. 42
- b. 210 ✓
- c. 212
- d. 208

$$s = \frac{12 + 35 + 37}{2} = 42. \text{ Area} = \sqrt{42(30)(7)(5)} = \sqrt{44100} = 210.$$

62 A triangular plot of land has sides 24, 45, 51. Find its area using Heron's formula.

a. 542

b. 60

c. 540 ✓

d. 538

$$s = \frac{24 + 45 + 51}{2} = 60. \text{ Area} = \sqrt{60(36)(15)(9)} = \sqrt{291600} = 540.$$

63 A triangular plot of land has sides 9, 12, 15. Find its area using Heron's formula.

a. 52

b. 56

c. 54 ✓

d. 18

$$s = \frac{9 + 12 + 15}{2} = 18. \text{ Area} = \sqrt{18(9)(6)(3)} = \sqrt{2916} = 54.$$

64 A triangular plot of land has sides 36, 105, 111. Find its area using Heron's formula.

a. 1890 ✓

b. 1888

c. 126

d. 1892

$$s = \frac{36 + 105 + 111}{2} = 126. \text{ Area} = \sqrt{126(90)(21)(15)} = \sqrt{3572100} = 1890.$$

65 A triangular plot of land has sides 21, 72, 75. Find its area using Heron's formula.

a. 758

b. 754

c. 84

d. 756 ✓

$$s = \frac{21 + 72 + 75}{2} = 84. \text{ Area} = \sqrt{84(63)(12)(9)} = \sqrt{571536} = 756.$$

66 Find the sum of the interior angles of a regular polygon with 20 sides, and the measure of each interior angle.

a. 3060, 162

b. 3240, 162 ✓

c. 3420, 157

d. 3240, 167

$$\text{Sum} = (20 - 2) \times 180^\circ = 3240^\circ. \text{ Each interior angle} = \frac{3240^\circ}{20} = 162^\circ.$$

67 Find the sum of the interior angles of a regular polygon with 72 sides, and the measure of each interior angle.

a. 12780, 170

b. 12600, 175 ✓

c. 12600, 180

d. 12420, 175

$$\text{Sum} = (72 - 2) \times 180^\circ = 12600^\circ. \text{ Each interior angle} = \frac{12600^\circ}{72} = 175^\circ.$$

- 68** Find the sum of the interior angles of a regular polygon with 4 sides, and the measure of each interior angle.

- a. 360, 95
- b. 180, 90
- c. 360, 90 ✓
- d. 540, 85

$$\text{Sum} = (4 - 2) \times 180^\circ = 360^\circ. \text{ Each interior angle} = \frac{360^\circ}{4} = 90^\circ.$$

- 69** Find the sum of the interior angles of a regular polygon with 30 sides, and the measure of each interior angle.

- a. 5220, 163
- b. 5040, 168 ✓
- c. 4860, 168
- d. 5040, 173

$$\text{Sum} = (30 - 2) \times 180^\circ = 5040^\circ. \text{ Each interior angle} = \frac{5040^\circ}{30} = 168^\circ.$$

- 70** Find the sum of the interior angles of a regular polygon with 6 sides, and the measure of each interior angle.

- a. 540, 120
- b. 900, 115
- c. 720, 120 ✓
- d. 720, 125

$$\text{Sum} = (6 - 2) \times 180^\circ = 720^\circ. \text{ Each interior angle} = \frac{720^\circ}{6} = 120^\circ.$$